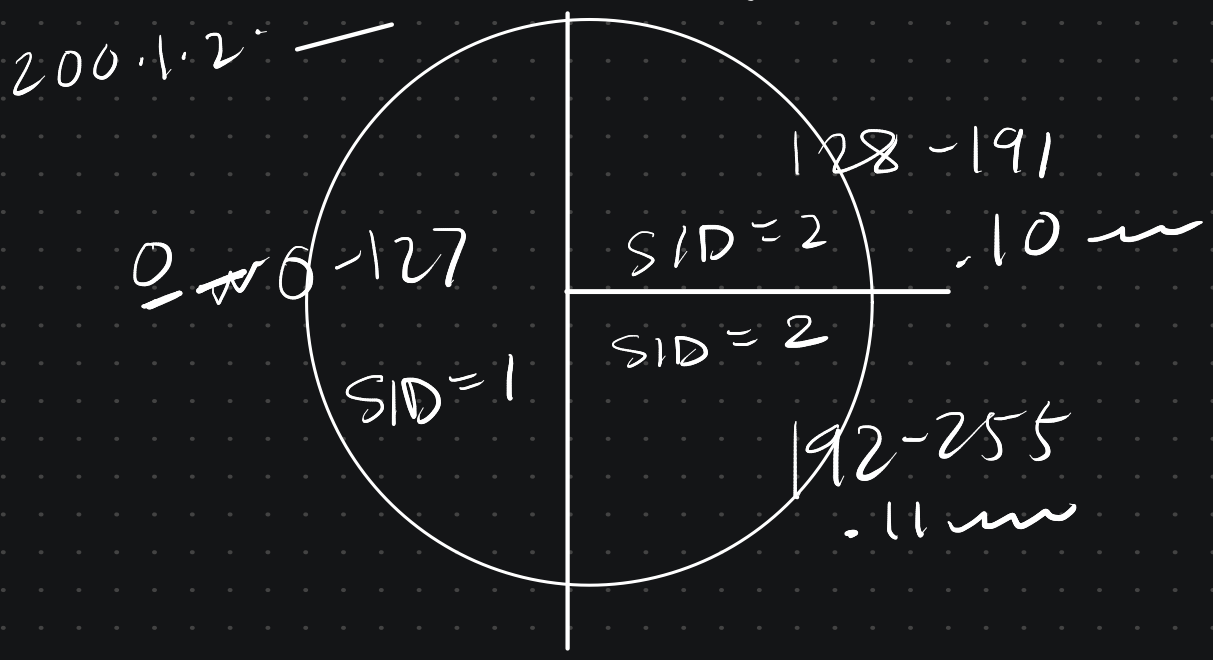
VLSM: Variable length Subnet masking.



Subnet mask for n/w 1:
255.255.255.128 [1 0000000]

sm for n/w 2: 255.255.255.192 [11 000000]

n/w 3: 255.255.255.192

As n/w sits A B & C are same, sm are same.

If n/w size is big, sm will be small.
↳ then hosts ↑ & HID ↑

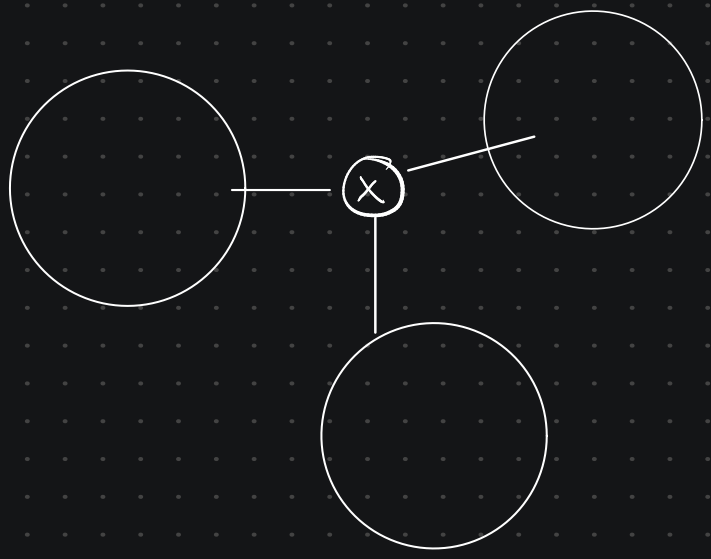
sm:



of 1's will be less if HID is more

routing table:

NID	SM	Interface
n	sm	a
n	sm	b
n	sm	c



10000000 → 128
11000000 → 192
11100000 → 224
11110000 → 240
11111000 → 248
11111100 → 252
11111110 → 254
11111111 → 255

In SM, # of 0's indicate HID.

255.255.255.192

of hosts = $2^6 - 2$

255.255.255.240 240 has 4 zeros

of hosts = $2^4 - 2$

255.255.248.0

248 has 3 zeros
0 has 8 zeros

of hosts = $2^{11} - 2$

255.240.0.0

4+8+8 zeros

of hosts = $2^{20} - 2$

Q. IP : 200.1.2.3.

SM : 255.255.255.192

of hosts / n/w = ?

→ $2^6 - 2$

Q. # of bits in its NID?

→ class C ∴ # of bits = 24.

Q. # of 1's in SM =

→ (NID + SID

→ 26 = NID + SID parts

26 = 24 + SID part

SID part = 2 bits.

Q. How many subnets are there?

→ $2^2 = 4$.

(don't subtract 2 for # of subnets)

Q. Say IP = 130.1.2.3

SM = 255.255.240.0

How many hosts/nw?

→ $2^{12} - 2$ 4+8 zeroes

Total # of 1's in SM?

→ 8+8+4 = 20

20 = NID + SID

20 = 16 + SID (NID in class B is 16 bits)

SID = 4 bits

∴ # of subnets = $2^4 = 16$

Q. IP = 100.1.2.3 SM = 255.255.0.0

→ In class A, NID = 8 bits

Total # of 1's = 16 = 8 + SID

SID = 8 bits

of subnets = $2^8 = 256$

Q: Divide a n/w into 2 parts. One of them into 2 halves. (Total 3 ⓘ)

→



or

(more practical, standard.

But there've been both type of questions in GATE.

---

classful addressing doesn't allow you to buy say 32 IP addresses. (min. 256 for class C)

Instead, there'll be no division beforehand.

like, instead of cutting the cake in shops beforehand, we give a knife to the customer (CIDR)

Here, the cake shop owner is IANA.

(Internet Assigned #s Authority)

ISPs buy IPs from IANA and us from ISPs

(there's rents and lease periods)

CIDR: Classless Inter-Domain Routing.

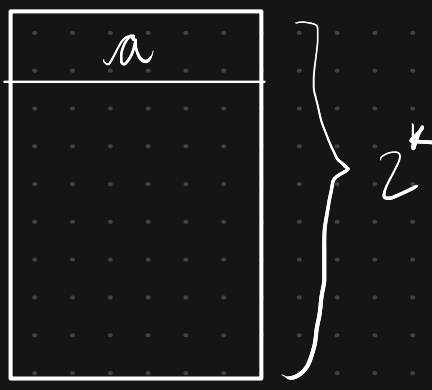
A piece is called a CIDR block.

Rules:

→ All IP addresses should be contiguous.

→ Size of a block is a power of 2. (2^n)

→ First IP adds. should be evenly divisible by the size of the block.



$$a \% 2^k = 0$$

Eg. Say 100.1.2.6 → 1st IP of block.

Size = 2^8

→

100.1.2.0 → 32-bit binary → Decimal.

Then divide that decimal by 2^8.

That's too dibficult.

Trick:

LSB of binary is remainder by 2.

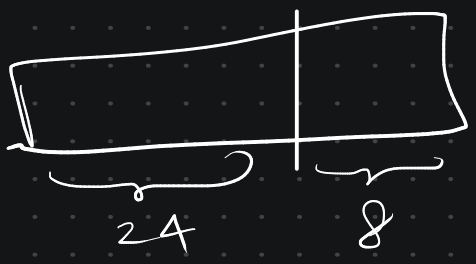
Ex. 11 % 2 → 1011 → 1.

11 % 2^3 → rem = 1011 i.e. 3

11 % 2^4 → 1011 → 4

So for an n-bit #,
Division by 2^k is the decimal equivalent of k LSBs.

e.g. 32 bits % 2^8
= last octet



Q. 10.1.2.32, size = 2^6.

→ rem = 00100000 → 32

rem ≠ 0. So CIDR isn't possible.
1st IP is not evenly divisible by size.

Q. 200.1.2.32, 2^{10}

→ Last 10 bits → 1000010000

i.e. 512+32 = 544
not evenly divisible.

Q. A block is given. Is it a CIDR block?

100.1.2.32
100.1.2.33
⋮
100.1.2.47

→ Rules:

1) contiguous ✓

2) size = 16 = 2^4 ✓ (47-32+1)

3) 1st IP = 100.1.2.32

last 4 bits → 0000

so divisible by size i.e. 2^4 ✓

So yes, it is a CIDR block.

NOTE:

block ID : first IP addr.

DBA: last IP

size → 2^4

∴ Host ID part is last 4 bits.

∴ # of IP address = 16

But # of hosts = 16 − 2 (one for NID/BID, other for DBA.)



* there is no classes A, B, C, D, E here. This is classless.

This n/w will be represented as

100.1.2.32 / 28

↑ NID/BID

In classful, we know NID & HID.

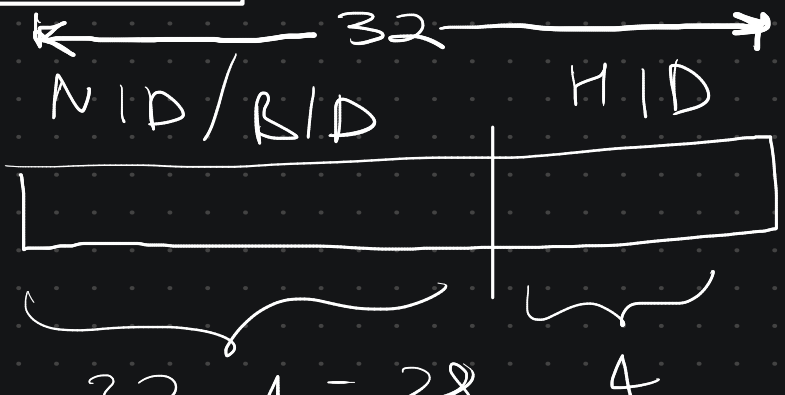
Here we don't.

So we write /28.

Same example repeated:

100.1.2.32
100.1.2.33
:
100.1.2.47

of IP addr. = 16
= 2^4.



← as # of IP addr. = 2^4.

To represent this n/w:

100.1.2.32/28 ← CIDR representation

Reason being that the division of # of bits is unclear.

Btw any IP addr. can be taken to repr. the block.

160.1.2.33/28

$$32 - 28 = 4$$

160.1.2.00100001

0010 0000 → 32

0010 0001 → 33

⋮ ⋮

0010 1111 → 47

So any of the IP addrs. can give the entire range by the # given by the slash.

e.g. 20.10.30.32 ⎫
20.10.30.33 ⎪
⋮ ⎪ ← satisfies CIDR rules
20.10.30.63 ⎭

→ # of IP addrs. = 32 = 2^5

∴ last 5 bits for HID.

32 − 5 = 27 bits for NID.

20.10.30.32/27

→ reverse process, say given

20.10.30.36/27

32 − 27 = 5

20.10.30.00010|00

0001	00000 ← 32
0001	00001 ← 33
⋮	
0001	11111 ← 63

e.g. 150.10.20.64
150.10.20.65
⋮
150.10.20.127

CIDR block or not?

→ size = 64 = 2^6 ✓ contiguous ✓

Last 6 bits = 000000 ✓

HID → 6 bits.

NID or BID → 32-6 = 26 bits.

150.10.20.64/26 ← CIDR block representation

1st IP is BID or NID or SID.

Last IP is DBA.

---

e.g. 100.1.2.20
100.1.2.21
⋮
100.1.2.30

→ # of IPs = 11 ≠ 2^k. NOT CIDR block.

Subnetting in CIDR:

say CIDR block is 20.1.2.3/24.

we'd like to divide into 2 equal parts.

24 is the NID/BID part.

HID is 8 bits.

NID	HID
20.1.2.	00000011

range:

00000000 → 20.1.2.0 → NID & BID

00000001 → 20.1.2.1

⋮ ⋮

11111111 → 20.1.2.255 → DBA

To divide into 2 parts:

20.1.2._ _ _ _ _ _ _ _

0 1

2^7 2^7

00000000 to 01111111

i.e.

20.1.2.0 to 20.1.2.127

10000000 to 11111111

i.e.

20.1.2.128 to 20.1.2.255

Btw /24 is kinda similar to Class C.

In original HID → 8 bits.

But now SID gets 1 bit.

new HID → 7 bits for either n/w.

So new NID/SID/BID → 32-7

: 25 bits

∴ Subnets will get /25.

Subnets are of same sizes.

↳ called FLSM in CIDR.

(fixed length subnet masking)

Q. 20.30.40.10/25 → Divide into 2 equal sized subnets.

→ HID is 32-25 = 7 bits

1 bit goes to SID.

So 20.30.40.0[0001010]

0001010 → 0

⋮

1111111 → 127

block is 20.30.40.0 to 20.30.40.127

NID

DBA.

To divide into 2 parts:

20.30.40.0 _ _ _ _ _ _ _ _

0 1

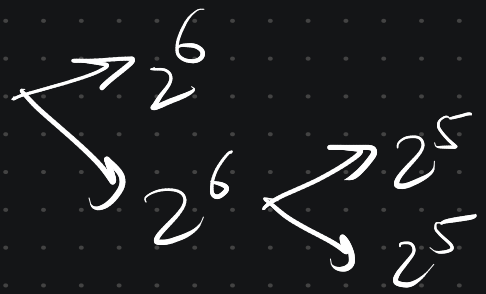
00000000
to
00111111
i.e.

20.30.40.0
to
20.30.40.63

20.30.40.64
to
20.30.40.127

Q. 20.30.40.10/25

Divide into 2 parts of sizes 2^6 & 2^6
& one of those into 2^5 & 2^5.

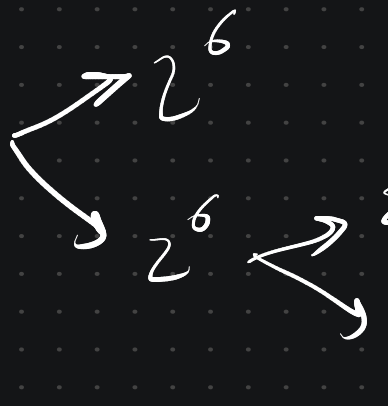


VLSM in CIDR

→ NID = 25 bits.

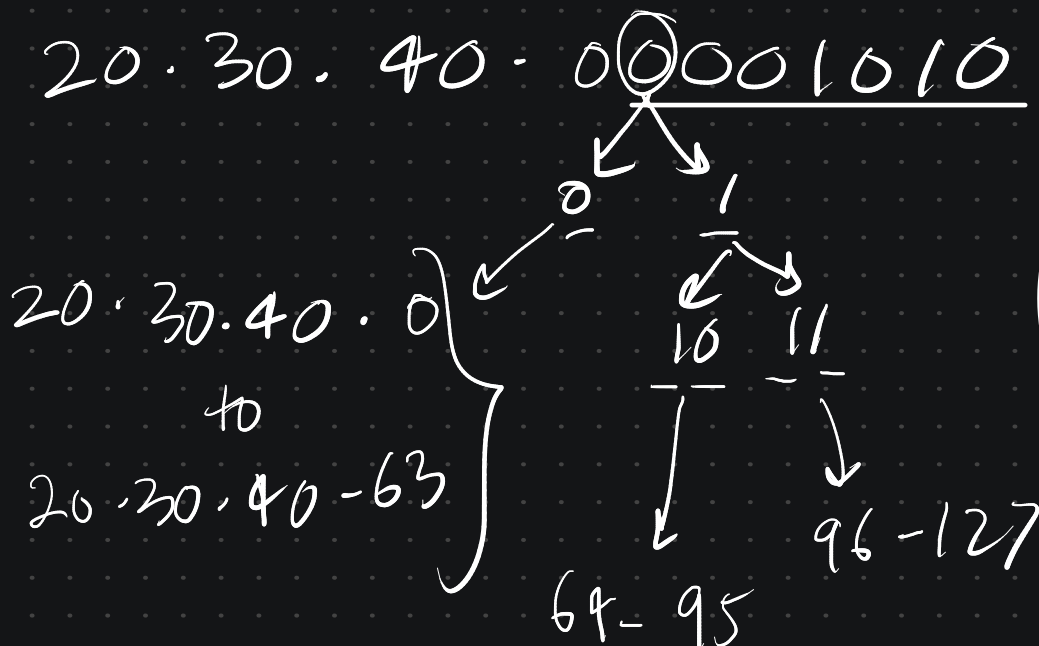
∴ HID = 7 bits.

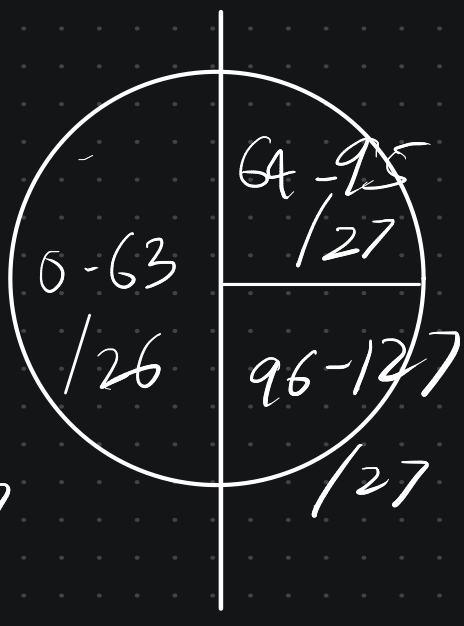
Size of n/w = 2^7



HID = 7 bits

20.30.40.00001010





Also, one more answer is there where the 0 part is divided into two 00 & 01 & 1

We have the IP address slash some x.
x bits are N/D. Don't touch those.

Choose 1st / 2nd bits whatever you want (based on # of subnets)

Then modify only the rest for the range.



say x+3
↑
will be new slash # for subnet

0000..
to
1111...

Q. 40.30.10.10/20.

Divide into 2^{11}

2^{11} → 2^{10}, 2^{10}

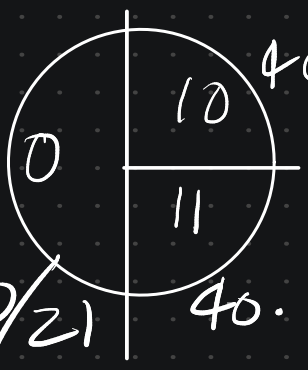
→ NID = 20 bits, HID = 12 bits.

Last 12 bits 1000001010

40.30.0000 1010.00001010

NID | BID 2^3 2^2

won't touch

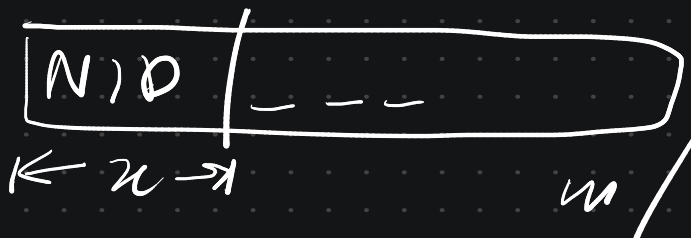


40.30.8.0/22

40.30.0.0/21 40.30.12.0/22

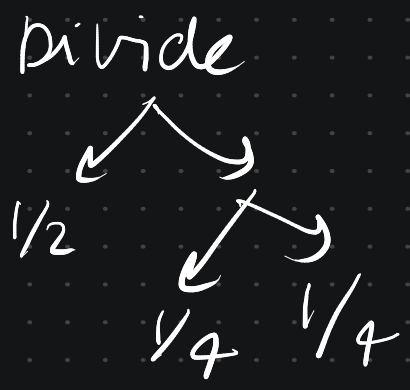
got from 20 bits + 1 bit of 0

20 + 1 1

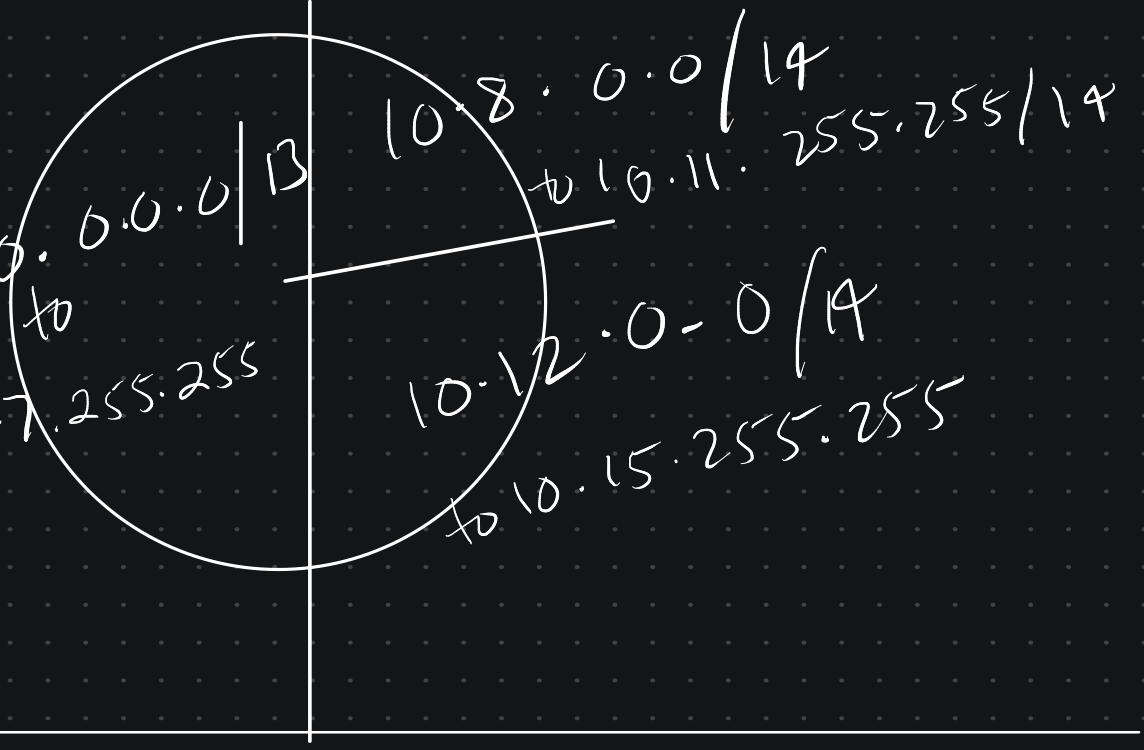


m/n+3

8. 10 · 10 · 10 · 10 / 12



Ans:

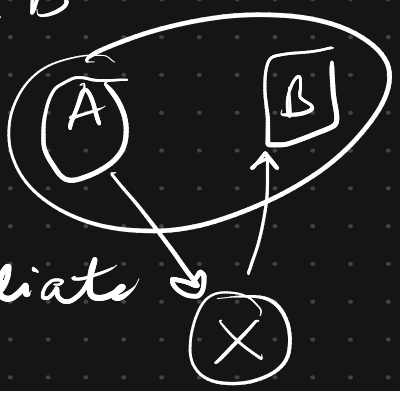


Say host A wants 2 send a msg to B.
2 cases: A & B are in same subnet
or diff subnet/n/w
via router.



How does A know where B is?
uses subnet mask.

sometimes the subnet mask creates an illusion that A & B are in diff n/ws.



router/gateway will act as intermediate

Say there's a host A with IP_A & SM_A.
Host B with IP_B, SM_B

4 possibilities:

At host A:

I_A		I_B	
B/W AND SM_A	AND	SM_B	← NID of B accd. to A
NID_{AA}		NID_{BA}	

If both are same, then accd. to A both are in the same n/w.
else router/gateway needed.

Ily, at B:

I_B		I_A	
SM_B		SM_B	
NID_{BB}		NID_{AB}	

ISP will give us the IP addresses.
What about n/w or subnet masks?
→ ISP or n/w admin (for private n/ws)

A can get any public IP address.
At A, we can't do IP_A AND SM_B as we don't know SM_B.

8. A: B:

$IP_A = 200.1.2.10$ $IP_B = 200.1.2.130$

$SM_A = 255.255.255.128$

→ Acc'd to A,

IP_B AND SM_A

11111111.11111111.11111111.10000000

AND 200. 1. 2. 10000010

200.1.2.128 ⇐ NID_{BA}

lly, 255.255.255.10000000

AND 200.1. 2. 00001010

200.1.2.0 ⇐ NID_{AA}

$NID_{AA} \neq NID_{BA}$

∴ Acc'd to A, B is in diff n/w.

Acc'd to A, a router should be used.

Q. $I_A = 200.1.2.10$ $I_B: 200.1.2.69$

$SM_A: 255.255.255.128$ $SM_B: 255.255.255.192$

Are these in the same n/w acc'd. to A & B?

→ $\frac{I_A}{S_A} = 200.1.2.0$ $\frac{I_B}{S_A} = 200.1.2.0$

∴ Acc'd. to A, both are in the same n/w.

$\frac{I_B}{S_B} = 200.1.2.64$ $\frac{I_A}{S_B} = 200.1.2.0$

∴ Acc'd. to B, both are in diff n/ws.

This discrepancy occurs due to diff. SMs.

What is DBA? All 1's in the HID part.

Q. 200.1.2.128/25 What is the DBA?

→ HID part has $32-25 = 7$ bits

200.1.2.10000000

(replace with 1's)

200.1.2.255 ← DBA.

NOTE: All HID part is 1's.
But all 1's may not be HID part.

Q. 200.1.15.255 ← DBA.
we don't know the NID part.
could you guess the NID part?

→

200.1.0000 1111.11111111

(this part COULD be the HID.

But we don't know how many of those 1's to consider.

Q. say we have 10 hosts. How many blocks do we buy?

→ we'll need 10+2 = 12 IP addr.
we'll get 16 (closest power of 2)
2^4.

∴ HID part = 4, NID part = 28

∴ we'd buy a /28 block.

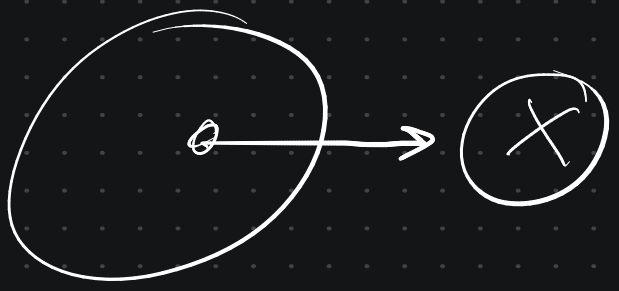
---

say 100 hosts

→ ∴ 102 IPs → 128 closest.

⇒ 7 HID part ∴ /25 block.

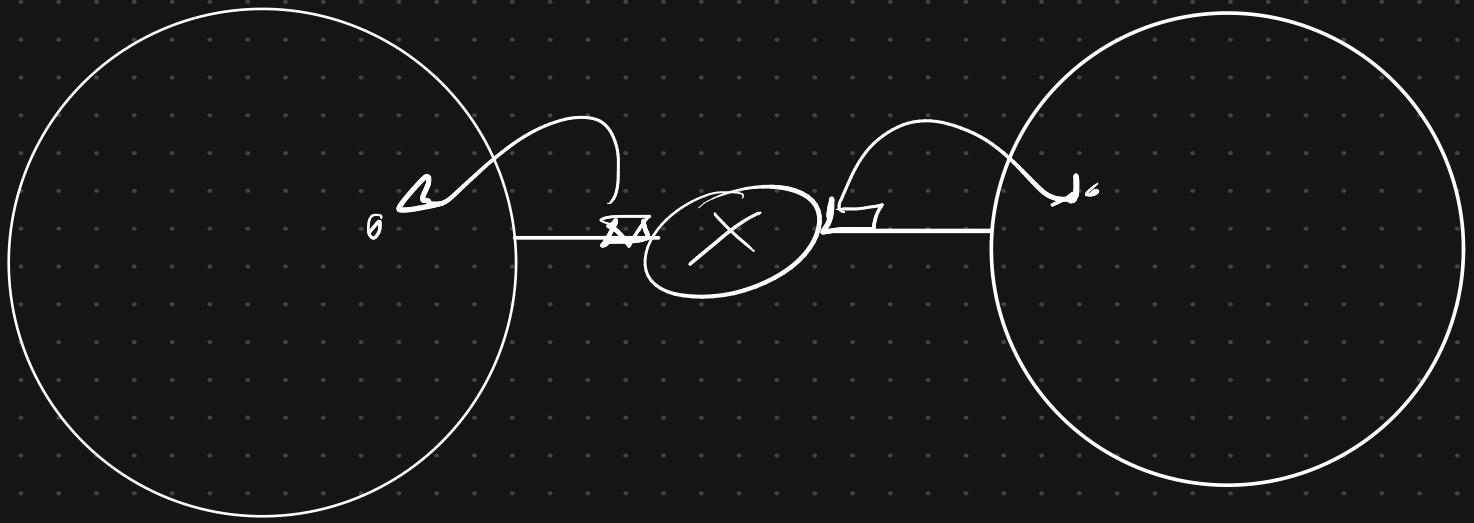
But say these hosts want to communicate, we need a router in the same n/w.
The router gets an IP addr. from the same network.



So # of IP addresses reqd increases by 1.

$$\#\text{ of IPs} = \#\text{ of hosts} + \underbrace{2}_{\text{NID \& DBA}} + \underbrace{1}_{\text{router}}$$

The router should be considered like a host.

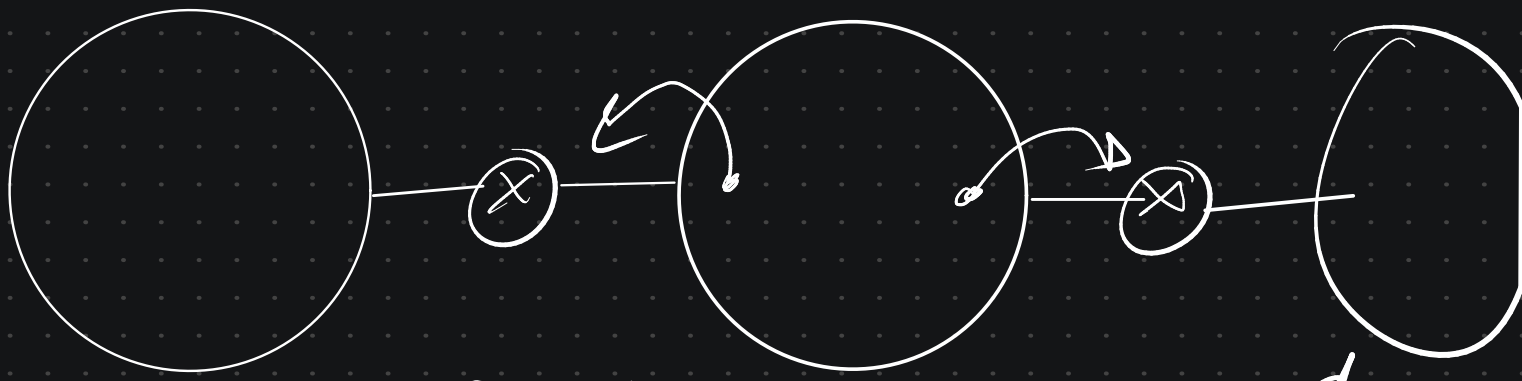


In a n/w like this, the router/interface will get one IP address each from either n/w.



we'd need a n/w here.

But no hosts.



For this subnet we need

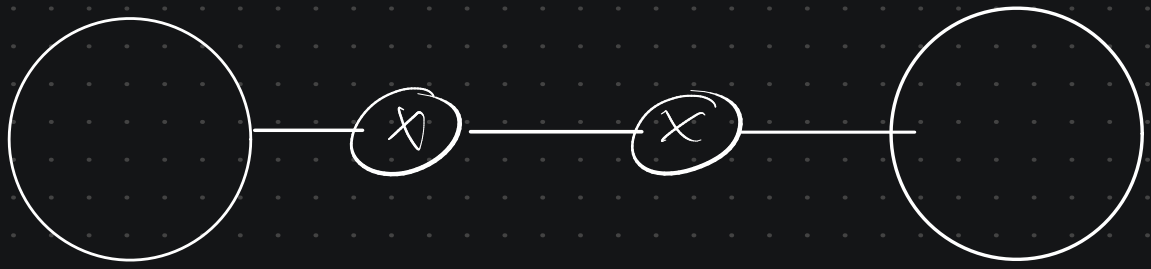
0 + 2 + 2 = 4 IP addresses for the middle n/w

0 ↑ no hosts in there

2 ↑ for either router

2 ↑ for NID and DBA.

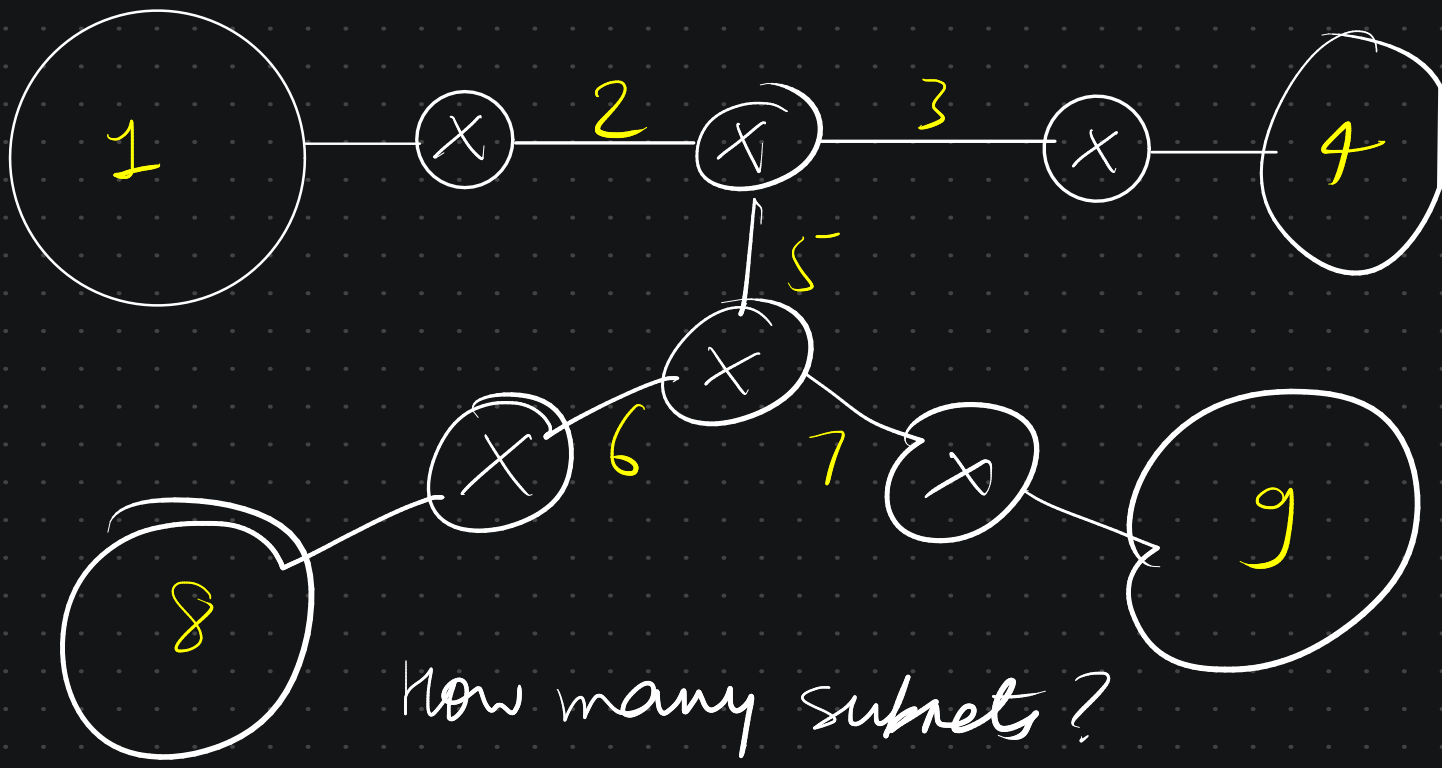
Q.



How many subnets are present?

→ 3.

Q.



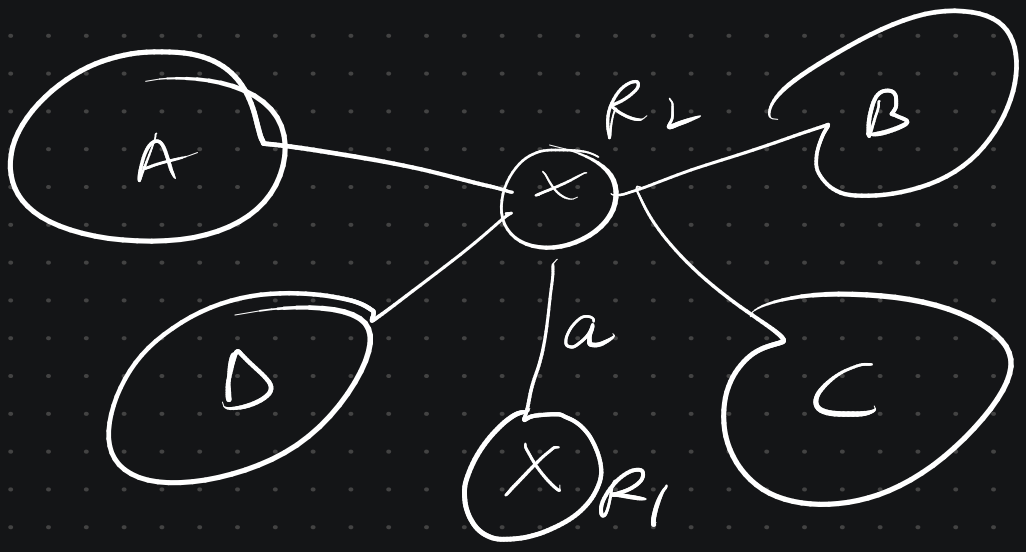
How many subnets?

→ 9

Same for gateways, switches, etc. in place of routers.

NOT bridges They're inside n/w's.

Only routers divide n/w's.



Routing table at R_1?

→

NID	SM	Next hop
N_A	S_A	a
N_B	S_B	a
N_C	S_C	a
N_D	S_D	a
0	0	Default

Problem: many entries going to the same interface.

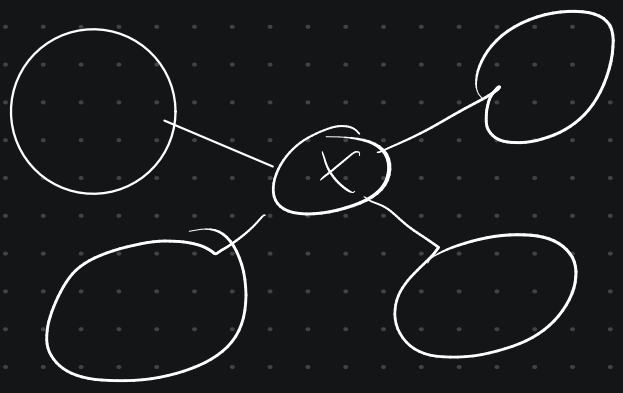
Why don't we assume the same n/w for all 4 and just have its details?

This is **supernetting**.

We combine the n/ws but NOT really. Just assuming so.



Supernetting



can we combine these?

→ NO

When can we combine?

→ only when we are able to give a supernet mask & a supernet ID.

Btw supernets & subnets are networks so SIDs will be called NIDs

SubID SuperID NID BID

consider

Rules for supernetting/Aggregation:

→ same as CIDR rules.

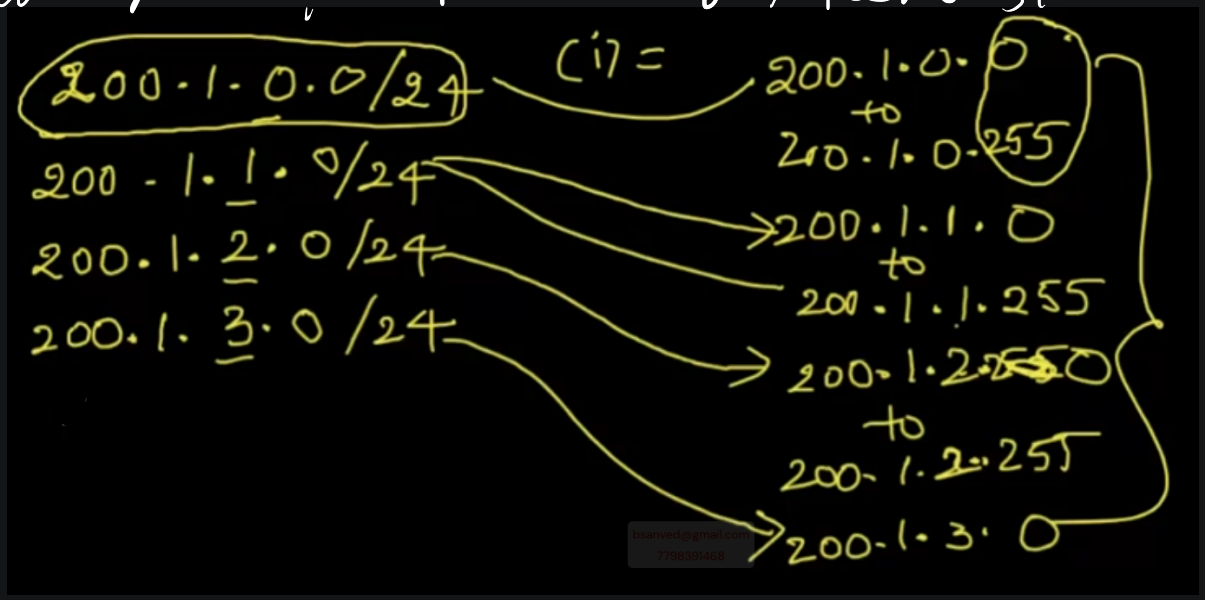
1) Contiguous IP addresses
2) same sized n/ws and total n/ws = 2^k
3) 1st IP of supernet should be evenly divisible by size of supernet.

Q. 200.1.0.0/24
200.1.1.0/24
200.1.2.0/24
200.1.3.0/24
} can we supernet these?

→ $4 = 2^2$ Also contiguous
last 2 bits must be 0
200.1.0.0 ✓
} YES

---

why contiguous? expand ranges, where one ends, next starts.



How do we get supernet mask?

→ In the fixed part, put all 1's.
In variable parts, put all 0's.

→ Simple method:
Total size is 2^{10}. ($2^8 \times 4$)
Size of each n/w is 2^8 & 4 n/ws.
HID → 10 bits.

∴ SID (supernet) part is 22.
so 200.1.0.0/22

Q. 200.1.32.0/24
200.1.33.0/24
⋮
200.1.47.0/24

→ # of n/ws = 16.
/24 is NID part. ∴ HID part = 32 − 24 = 8
size of each n/w = 2^8.
total size = $16 \times 2^8 = 2^{12}$
∴ 2^{12} is # of IPs

1st IP

200.1.00100000.00000000

Least 12 bits should be zero. They are ✓

∴ they can be combined to supernet.

Supernet ID is 200.1.32.0

Total size is 2^{12}.

∴ HID part is 12 bits

⇒ NID part is 20 bits.

∴ SM is /20 i.e. n/w mask has 20 1's.

11111111.11111111.11110000.00000000

i.e. 255.255.240.0

200.1.32.0/20

8. 100.1.2.0/25
100.1.2.128/26
100.1.2.192/26

→ 2nd and 3rd n/ws can be aggregated.

100.1.2.10000000] 100.1.2.128/26
⋮
10111111

100.1.2.11000000] 100.1.2.192/26
⋮
11111111

$2^6 \times 2 = 2^7$ ∴ HID part = 7
NID part = 25

↑ size of each ↑ total size

So we get 100.1.2.128/25

Now 1st one & this can be combined.

100.1.2.0/25 → [100.1.2.00000000
⋮
01111111

100.1.2.128/25 → [100.1.2.10000000
⋮
11111111

20 $2^7 \times 2 \longrightarrow 2^8$

$\therefore$ NID $\rightarrow$ 24 bits

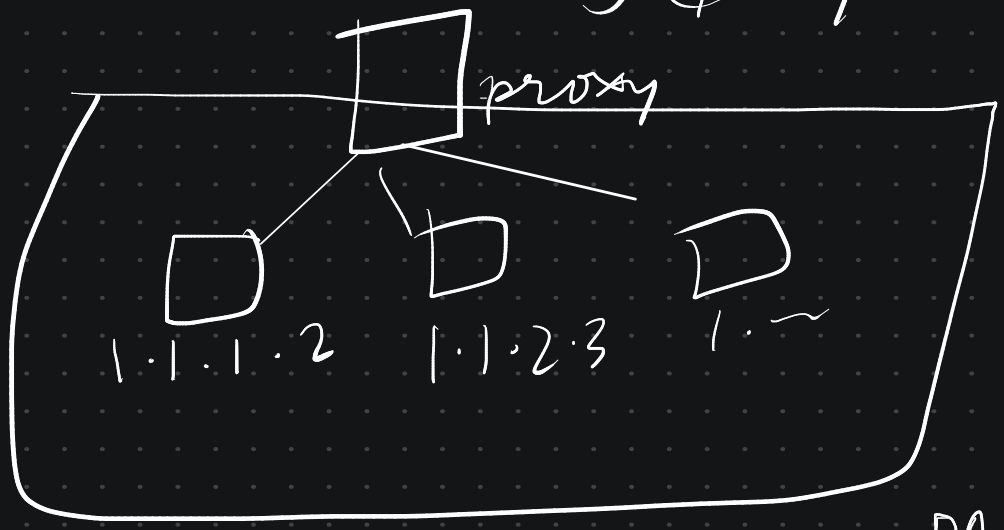
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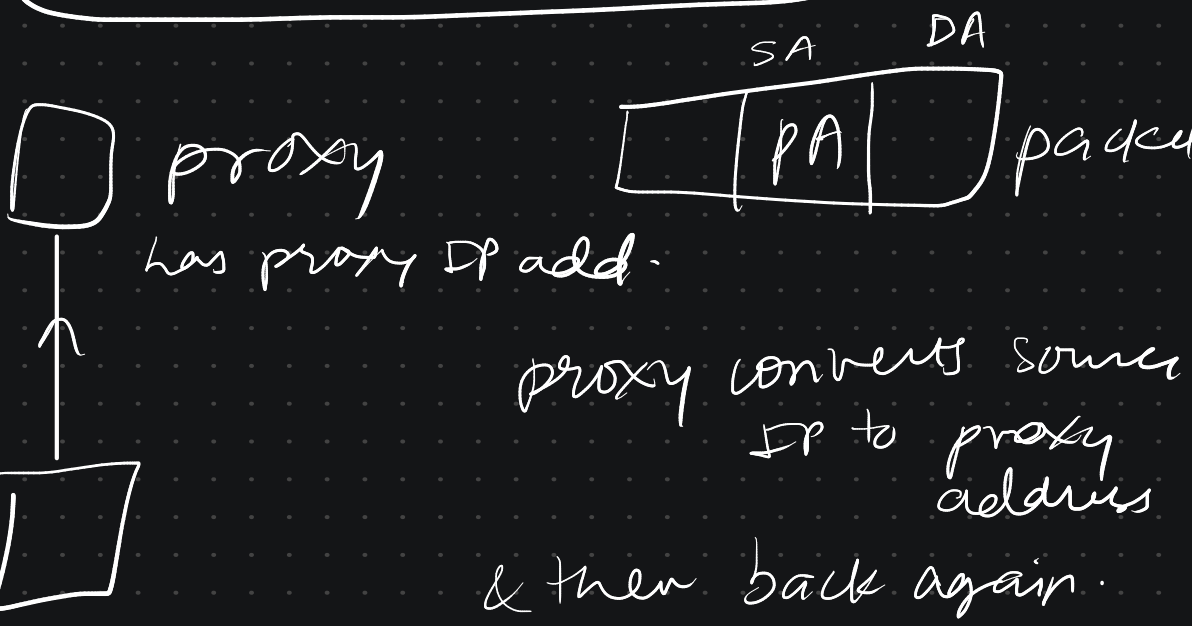
you'll just have to observe

$$\left.\begin{matrix}26\\26\end{matrix}\right\} \left.\begin{matrix}25\\25\end{matrix}\right\} 24$$

NAT

In n/w s say like in a college, the college won't buy IPs for all n/w, they'll buy 1 for gateway (or mostly proxy server)





This is Network Address Translation.
IP addresses are reserved that can be used for free in local n/w s. [private addresses]
valid only in private n/w s.

10.0.0.0 to 10.255.255.255
172.16.0.0 to 172.31.255.255
192.168.0.0 to 192.168.255.255
} ranges

So the college would buy only one IP address & use these ranges.
Outside, the proxy server IP would be visible.
